

Modeling Human Responses to Multimodal AI Content (Supplementary Material)

This supplementary material provides additional experimental details, analyses, and examples to complement the main paper.

1 Construction of MhAIM Dataset

Here we provide additional details on the construction of the proposed Multi-modal Human & AI Generated Media Dataset (MhAIM).

1.1 Stimuli Collection and Generation

Our MhAIM Dataset includes stimuli coming from diverse sources. Table S1 reports the detailed sources of the MhAIM Dataset. Our AI images are generated based on the text from the sources; our AI text are generated based on collected images or AI images.

1.2 AI Headline Generator and AI Content Generation

We generate AI content based on the real-world data. For AI images, we used texts from our initial collection as prompts for the Stable Diffusion XL 1.0 [13] to generate corresponding images. For AI texts, we built our own AI Headline Generator to generate news headlines, social media posts, and phishing messages. The generator is powered by the large language model ChatGPT-3.5 [14], serving as its core cognitive engine. This generator is further equipped with a suite of specialized tools to facilitate content creation: an Image Captioning Tool [15] and a Video Captioning Tool [16] to discern visual content, a VQA (Visual Question Answering) Tool [15] to retrieve specific details from images. Besides, we provide the AI headline Generator with a News Headline Generator Tool, a Social Media Post Generator Tool, and a Phishing Message Generator Tool. The News Headline Generator Tool has versatility to mimic 19 distinct styles from major outlets like BBC, CNN, and Fox News, and the capability to imbue generated headlines with 16 unique attributes such as “eye-catching”, “breaking”, and “shocking”. The Social

Media Post Generator Tool can emulate 11 different social media styles including Twitter, Instagram, and Facebook, and can impart 24 diverse attributes to the generated content, ranging from “interesting” to “exciting”. Finally, the Phishing Message Generator Tool simulates 5 different phishing messages as those appear in our daily social media platforms and mail boxes.

When input with a prompt with an image, the generator’s workflow begins by discerning the need for a tool, for instance, it might employ the Image Captioning Tool to understand the image’s content. Optionally, it could deploy the VQA Tool for added context. The chain of thought of the AI Headline Generator is illustrated in Fig. S1. Tables S2 to S6 delineate the platform styles and attributes utilized during the creation of news headlines, social media posts, and phishing messages, respectively. During the generation process, our generator randomly selects the attributes and styles, depending on the type of content (news headline, social media post, phishing message) being created. Fig. S2 shows some consolidated examples from the final MhAIM Dataset.

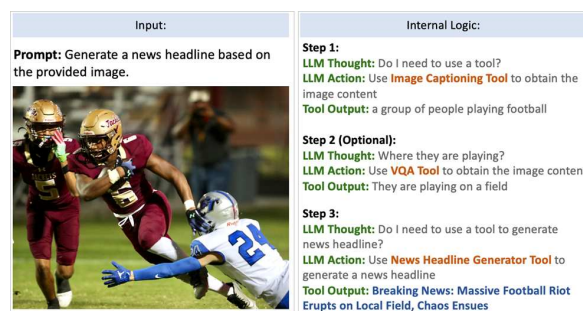


FIGURE S1: We created an application called the AI Headline Generator, designed to produce AI-generated headlines for building the MhAIM Dataset. The figure depicts the chain of thought of the Generator.

1.3 Human Psychophysics Study and Data Annotation

We conducted a comprehensive human study on a subset of MhAIM. Participants engaged with a thorough questionnaire probing their reactions, con-

TABLE S1: The initial collection of MhAIM Dataset is comprised of data from diverse sources.

Source	Modality	Label type	No.
TrueFact_FND [1]	text	authenticity	20,983
N24News [2]	text	authenticity	1,769
Emod [3]	image	sentiment	1,018
NVED [4]	video	sentiment	404
Moments in time [5]	video	action	3,999
DFW [6]	video	deepfake	1,057
FaceForensics++ [7]	video	authenticity	1,109
FVC [8]	text, video	authenticity	359
YouTubeAudit [9]	text, video	authenticity	477
Fakeddit [10]	text, image	authenticity	10,000
COSMOS [11]	text, image	consistency	359
MVSA [12]	text, image	consistency	300
Snopes (own)	text, image	authenticity	1,475
Total			43,309

TABLE S2: Platform styles used in news headline generation by our AI Headline Generator.

Style categories	News platforms	Has political bias?
General news channels based in US	CNN News, Fox News Channels, ABC News, NBC News, The New York Times, NPR News	Y
General news channels based in UK	BBC News, Reuters, The Guardian, The Daily Telegraph	Y
General news channels based in China	China daily	Y
General news channels based in companies	Google News (skewed left based on Allsides.com)	Y
Independent, nonprofit general news organization	The Real News Network	N
Business and economic-focused news channels	The Economist, The Wall Street Journal, Bloomberg	Y
Technology focused news channels	TechCrunch, ScienceDaily	N
Satirical digital media	The Poke (from UK), The Onion (from US, lean left)	Unsure

TABLE S3: Attributes for news headlines used in our AI Headline Generator.

Categories	Attributes
Functional attributes	misleading, fake, debunking, promoting, alarming, sarcastic
Relating to impact	go viral, controversial, significant, noteworthy, eye-catching
Relating to emotions	breaking, shocking, sensational, interesting, intriguing

victions, and perspectives concerning the presented single- or multi-modal information. We built our own

study platform. Fig. S3 shows the user interface of our human study.

Owing to the significant impact of view size on human perception and sensitivity to visuals [17], [18], [19], [20], we excluded submissions completed on a screen size smaller than 1001×563 pixels. We aimed to simulate scenarios most common to desktop or laptop usage, typically featuring screen sizes larger than 1001×563 pixels. This filtration resulted in a final participant sample of 375 from the US, 176 from India, and 180 from Singapore. To ensure the quality of human data, we further excluded audio and videos exceeding five seconds in length.

Participants from the US, India, and Singapore were chosen to incorporate a diverse range of per-



FIGURE S2: Examples from the MhAIM Dataset showing origins of text (T) and visual (V), authenticity (AUTH), and average human perceptions. Bottom entries in the first two columns derive from their above entries. Red outlines denote full AIGC (both text and visuals AI-generated), orange for AI text, blue for AI visuals (visualized in subsequent figures), and green for entirely human-crafted content. Unless noted, AI texts and visuals are generated by ChatGPT and Stable Diffusion, respectively.

TABLE S4: Responses collected in our human study.

Type	Detailed Attributes
Emotional Responses	Happy? Sad? Angry? Frightened? Impressed?
Behavioural Reactions	Will you share the content with friends? Do you believe this content?
Perception Related to AI	Is this computer-generated? Why?
Multimodal Consistency	Text matches the visual content? Textual sentiment matches the visual sentiment? Are objects mentioned in the text shown visually?
Genre	Genre of the news/social media post?

spectives in the study, considering the significant variances in media consumption, digital literacy, and cultural contexts across these countries. The US often serves as a pivotal point for technological advancements and its populace is immersed in a dynamic media environment. India offers insight into a massive, diverse user base navigating a multifaceted media landscape, and Singapore provides a perspective from a technologically advanced Asian nation with stringent media regulations. This geographical and cultural diversity enriches the data, providing a broad spectrum of responses to AI-generated content, which is crucial for understanding global patterns in information perception and sharing across different socio-cultural environments.

TABLE S5: Attributes and platform styles for social media posts generation by our AI Headline Generator.

Categories	Attributes/styles
Functional attributes	breaking news, misleading, debunking, promoting, sarcastic, thought-provoking, informative, insightful, alarming
Relating to impact	viral, controversial, eye-catching, compelling, trending, influential
Relating to emotions	shocking, sensational, interesting, intriguing, hilarious, inspiring, memorable, heartwarming, funny, intriguing
Platform styles	Twitter, Instagram, Facebook, TikTok, Reddit, YouTube, WhatsApp, Pinterest, LinkedIn, Snapchat, WeChat

TABLE S6: Attributes and platform styles for phishing message generation by our AI Headline Generator.

Categories	Attributes/styles
Functional attributes	compelling, noteworthy, eye-catching, sensational, alarming, interesting, intriguing, debunking, promoting
Platform styles	Twitter, Instagram, Facebook, TikTok, Reddit, YouTube, WhatsApp, Pinterest, LinkedIn, Snapchat, WeChat, Email

Fig. S4 illustrates the demographics of our human study participants. The comprehensive questionnaire utilized in our study can be found at the end of this Appendix.

Please answer the following questions about the post on the left.

T1. Please let us know how your feelings about the post on the left:

	1	2	3	4	5	6	7
1. How happy does the post make you?	☐	☐	☐	☐	☐	☐	☐
2. How sad does the post make you?	☐	☐	☐	☐	☐	☐	☐
3. How angry does the post make you?	☐	☐	☐	☐	☐	☐	☐
4. How frightened does the post make you?	☐	☐	☐	☐	☐	☐	☐
5. How impressive do you find this post?	☐	☐	☐	☐	☐	☐	☐

T2. How likely will you share this post with others, either through verbal communication, reporting on your social media, or any other means?

	Very Unlikely	Unlikely	Neutral	Likely	Very Likely
T2. How likely will you share this post with others, either through verbal communication, reporting on your social media, or any other means?	☐	☐	☐	☐	☐

T3. How trustworthy is this post?

	Do not at all trustworthiness	Untrustworthy	Neutral	Trustworthy	Very trustworthy
T3. How trustworthy is this post?	☐	☐	☐	☐	☐

T4. How likely do you think this post is computer generated (i.e., either text or visual content or both are created using computer software or AI algorithms)?

	Very Unlikely	Unlikely	Neutral	Likely	Very Likely
T4. How likely do you think this post is computer generated (i.e., either text or visual content or both are created using computer software or AI algorithms)?	☐	☐	☐	☐	☐

FIGURE S3: User interface of our human study.

1.4 An Overview of Human-Annotated Data

Here we report item counts per category in the annotated MhAIM subset: i) Non-AIGC true information (2058), ii) Non-AIGC misinformation (1071), iii) Visual-only AIGC (1532), iv) Text-only AIGC (1606), and v) Full AIGC (2685). Table S7 reports the distribution across annotated genres and origins. Human-crafted true information was rated by participants as being the most consistent, followed by human-crafted misinformation, while AIGC, particularly with both AI-generated text and visuals, was least consistent ($F_s \geq 90.09$, $p_s \leq .0001$). The results are visualized in Fig. S5. Due to the small sample size of phishing messages ($n = 24$) in the rated subset, we categorized them as social media posts throughout this paper.

2 Human Perception of AIGC

Additional Observation 1: Human sensitivity to AIGC is influenced by individual factors such as gender, nationality, and prior experiences.

As an exploratory analysis (not based on pre-registered hypotheses), we explored the impact of participants' demographic backgrounds on their responses. ANOVA results suggested that Singaporean participants exhibited marginally higher sensitivity compared to their Indian counterparts ($p = 0.033$, Singaporean: $d' = 1.46$; Indian: $d' = 0.96$). Females displayed a notably conservative bias, leaning more towards classifying content as human-crafted than males did (independent samples t -test, $t(706) = 3.13$, $p = 0.002$; males: $c' = 0.92$; females: $c' = 1.24$). While prior experience with AIGC did not alter sensitivity,

it significantly influenced bias (F values ≥ 7.41 , p values ≤ 0.0001). Specifically, those familiar with AI-generated tools such as ChatGPT were more inclined to classify content as AIGC (p values ≤ 0.001).

Additional Observation 2: Participants were more accurate in identifying AIGC on news genres that they are most familiar with.

Participants exhibited varied judgments accuracy across different news genres ($F_s \geq 5.06$, $p_s \leq .0001$). It appears that they identified AIGC more accurately in genres with which they were familiar, such as entertainment and sports news, while their accuracy declined for less familiar content, such as news about disasters and accidents, finance, and economics ($p_s \leq .0001$). Fig. S6 presents the variations in judgment across different genres.

Additional Observation 3: Content eliciting significant negative emotions often correlates with high perceived AIGC likelihood.

We explored human behavior in AIGC detection relative to five emotions: happiness, sadness, anger, fear, and impressiveness. While sensitivity d' was stable across emotions ($F_s \leq 2.53$, $p_s \geq .020$), bias c' turned negative with significant negative emotions ($F_s \geq 4.53$, $p_s \leq .0002$), indicating a tendency to rate such content as AIGC (see Fig. S7).

Additional Observation 4: Distinct perceptual patterns were identified in participant responses to AIGC versus human-crafted multi-modal content.

Here we conducted a thorough analysis to understand the nuanced impacts of AI-generated content on human perception, focusing on variances across five types of information. Owing to data constraints on hit rates and false alarm rates, we utilized judgment accuracy for this exploration instead of Signal Detection Theory (SDT), probing the fluctuations in human judgment accuracy associated with perceived multi-modal consistency.

The perception of human sentiments revealed noticeable disparities when comparing multi-modal AI-generated content with human-crafted content ($F_s \geq 3.81$, $p_s \leq 0.0002$). Pertaining to judgment accuracy, patterns diverged based on multi-modal consistency. For human-crafted content, judgment accuracy showcased an approximately linear relationship, varying with the degree of content and sentiment consistency ($F_s \geq 5.16$, $p_s \leq 0.00002$). Conversely, for AIGC, accuracy dipped notably at medium consistency levels, resulting in a valley-shaped curve ($F_s \geq 4.72$, $p_s \leq 0.0001$).

Fig. S8 illustrates the significant behavioral disparities between human-crafted (either true information or misinformation) and AI-generated information.

TABLE S7: Distribution of genres and origins for news and social media posts in the annotated MhAIM subset.

Medium	Genre	Non-AIGC true info	Non-AIGC mis-info	Visual-only AIGC	Text-only AIGC	Full AIGC
News	Politics	25	70	94	80	102
	Health & life science	22	24	58	48	84
	Food & food industry	3	18	19	25	35
	Entertainment	55	84	140	172	195
	Sports	31	27	55	72	97
	Finance & economics	17	41	51	66	94
	Disasters & accidents	10	33	42	44	91
	Other sciences & technology	10	18	31	62	74
Social media post	User-generated daily life content	48	2	63	55	152
	Pop culture, memes & humor	22	5	63	34	80
	Advertisement	15	0	12	19	34
	Political news	47	2	57	15	53
	Health, science & food news	14	0	18	28	29

Judgment accuracy for human-crafted true information remained somewhat stable, with a subtle upward trend as both content and sentiment consistency amplified ($F_s \leq 2.31$, $p_s \geq 0.031$). Judgment accuracy for human-crafted misinformation exhibited a similar, yet more accentuated, increasing trend ($F_s \geq 5.16$, $p_s \leq 0.00003$). In contrast, for AI-generated content, participants demonstrated notably lower accuracy when the content displayed medium consistency in both content and sentiment ($F_s \geq 4.72$, $p_s \leq 0.0001$).

Observation 5: Participants' emotional responses to visual content differed depending on its perceived origin (AI-generated or human-crafted). They had most negative feelings when unable to distinguish whether the content was AI-generated or human-made, resonating with the "uncanny valley" theory.

Emotional perceptions varied with perceived AIGC likelihood. For visual content, participants exhibited the strongest negative sentiments when uncertain of its origin ($F \geq 8.07$, $p \leq .0001$), aligning with the *uncanny valley* theory [21] (see Fig. S10).

Additional Observation 6: Information receptivity is influenced by its type and origin.

Information trustworthiness and impact fluctuate based on its type and origin. For AIGC, single-modal social media texts retain the highest values across both attributes, succeeded by news headline text, whereas multi-modal information generally exhibits lower trustworthiness and impact ($F \geq 10.91$, $p \leq 0.0001$); see Fig. S11 (a). Conversely, human-crafted information reaches peak values for multi-modal news content; refer to Fig. S11 (b). Human-crafted materials consistently exceeded AI-generated content across both

attributes, as illustrated in Fig. S12.

Furthermore, posts seen as AI-generated but frequently shared showed higher text-visual consistency, elicit stronger emotions, and receive greater trustworthiness than those appear AI-generated but less often shared, as illustrated in Fig. S13.

3 Predicting Human Receptivity

Fig. S14 illustrates the prompt utilized in T-Lens for asking LLMs (GPT-4o; Gemini) to produce explanations for the predicted outcomes. A screenshot of T-Lens is shown in Fig. S15. We compare the explanation of T-Lens with predictions from GPT-4o and Gemini. As illustrated in Fig. S16, all responses collectively underscore the challenge posed by the disconnect between the headline's serious environmental message and the seemingly unrelated image, impacting the post's credibility and viral potential. Compared to the responses from GPT-4o and Gemini, T-Lens offers a more analytical and structured approach, taking into account the consistency in sentiment between text and image. It offers a logical and comprehensive assessment of its low viral potential due to perceived in-authenticity, which originated with mismatch in both content and emotion between text and image. Equipped with prior knowledge from our human study, T-Lens emphasizes credibility. It establishes a logical link between trust and dissemination propensity. This suggests a more cautious and discerning user engagement. It helps in understanding the dynamics of human responses in the context of AI media's evolving landscape. However, GPT-4o discusses individual differences that are not accounted for in T-Lens. Our future work involves in-



FIGURE S4: Our participants have a diverse demographic background in terms of age, race, education, country of residence, and prior AIGC-related experience.

corporating these human variances. The prompt used for GPT-4o and Gemini is shown in Fig. S14.

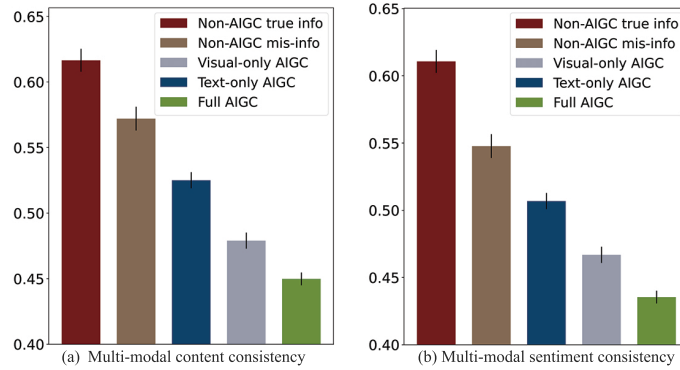


FIGURE S5: Content (a) and sentiment (b) consistency in multi-modal content is highest for human-crafted authentic information and lowest when both text and visual elements are AI-generated. In all figures throughout this document, error bars denote standard errors.

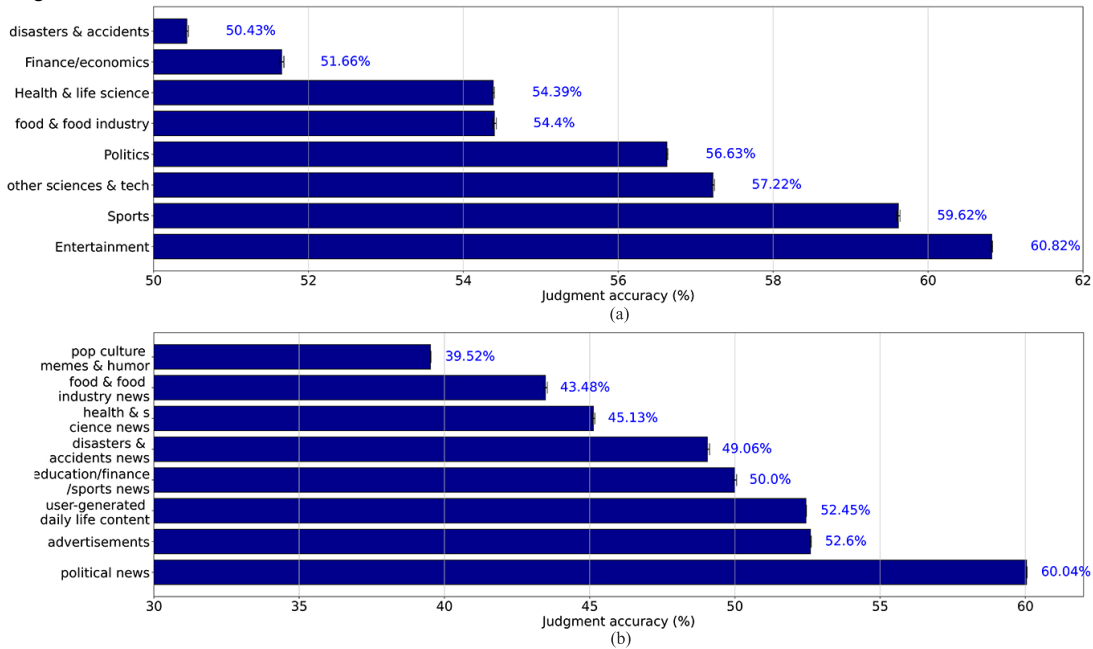


FIGURE S6: Participants' judgment accuracy differs on different news (a) and social media (b) genres.

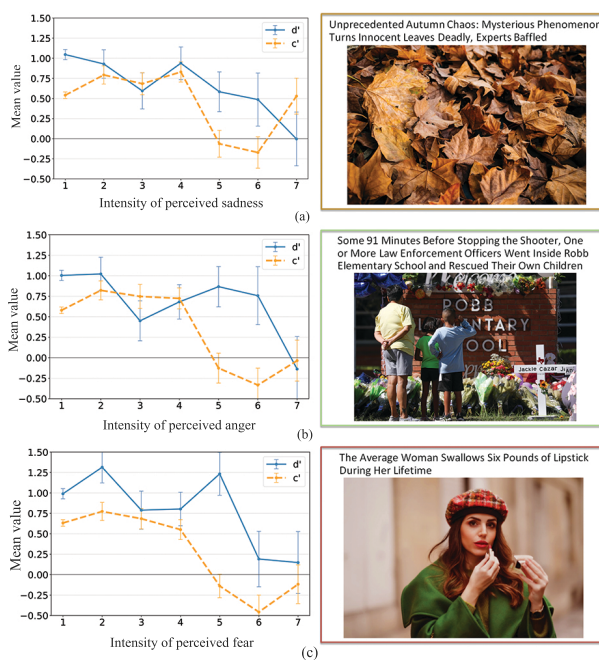


FIGURE S7: Negative emotions like sadness (a), anger (b), and fear (c) correlate with a higher tendency to judge content as AIGC. The right column shows corresponding content that evokes intense emotions (6 out of 7 on the 7-scale Likert scale). In all figures, the error bars represented standard errors.

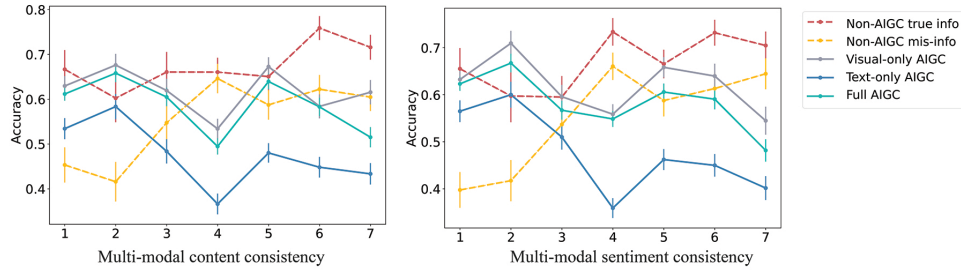


FIGURE S8: Human judgment accuracy of AIGC varies between human-crafted and AIGC content. For human-crafted content (depicted by red and yellow dotted lines), judgment accuracy appears roughly linear with varying levels of content consistency (left) and sentiment consistency (right). In contrast, for AIGC content (represented by gray, green, and blue solid lines), accuracy dips noticeably at medium consistency levels, resulting in a valley-shaped curve. “Non-AIGC” means human-crafted. “Visual-only AIGC” means only the visual component in the information is generated by AI. “Text-only AIGC” means only the textual component in the information is AI-generated.

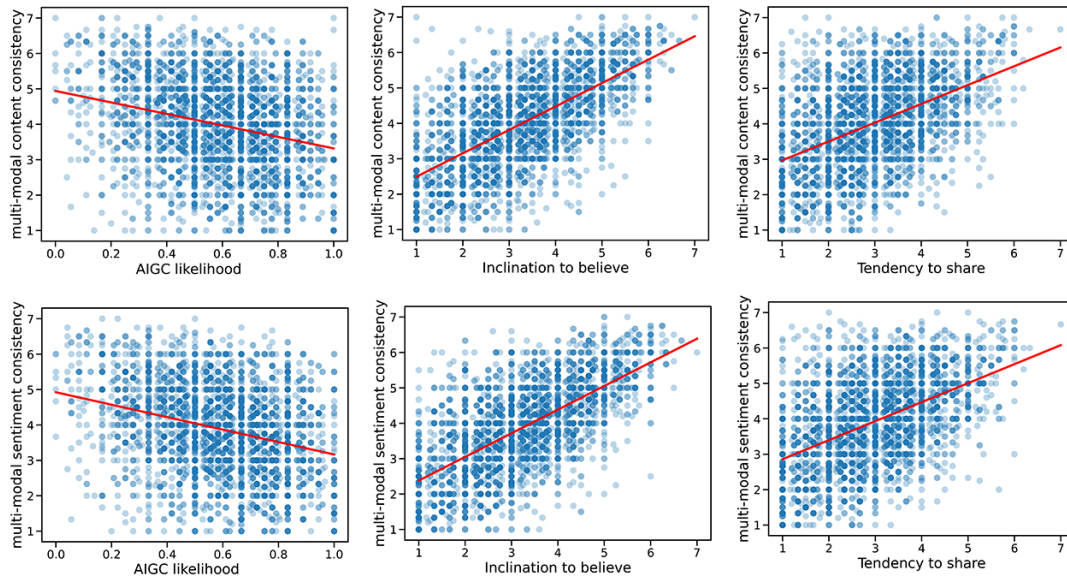


FIGURE S9: Correlations between content (top row) and sentiment (bottom row) consistency and participants’ ratings of likelihood of AIGC and inclination to share and believe.

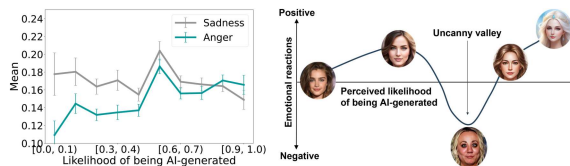


FIGURE S10: Emotional responses to AI-generated visuals vary based on their perceived AI origin, with negative sentiments highest for ambiguous sources (a), reflecting the Uncanny Valley Theory [21] in the context of AI imagery (b).

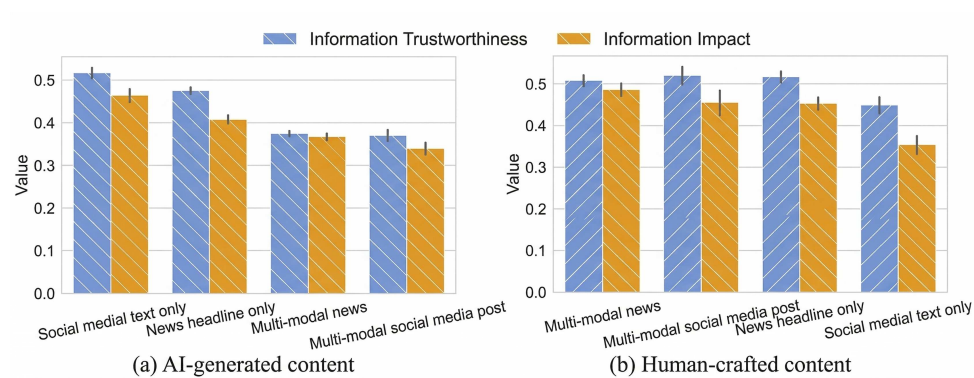


FIGURE S11: Information trustworthiness and impact vary based on origin, type, and modality. In our data, text-only content generally attains higher values than multi-modal content across both attributes for AIGC (a). In contrast, for human-crafted content, multi-modal news consistently achieves the highest overall values (b).



FIGURE S12: Human-crafted information typically shows higher trustworthiness (a) and impact (b) than AI-generated content, while AI-generated text consistently surpasses AI-visuals and full AIGC. Below each bar graph, sample AIGC content with top scores ($> .90$) is displayed.

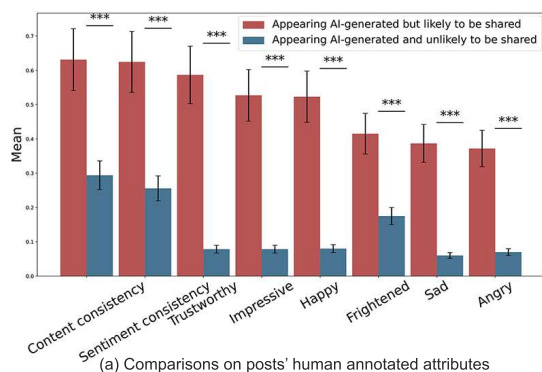


FIGURE S13: (a) Posts seen as AI-generated but frequently shared showed higher text-visual consistency, elicit stronger emotions, and receive greater trustworthiness than those less often shared. (b) Examples of posts likely perceived as AI-generated yet likely to be shared. (c) Examples of posts perceived as AI-generated and less likely to be shared.

Code block

```

1  You are a ReAct Agent assistant help user to answer questions about News. You will be
  given a user question and images (optional). You need to complete the task with
  provided tools.
2
3  ## User Question
4  <user_question>
5  {{ user_question }}
6
7  ## News Title or Text
8  <news_text>
9  {{ news_text }}
10 </news_text>
11
12 {% if news_image %}
13 ## News Image
14 Image Path: {{ news_image }}
15
16 <image_placeholder>
17 {% endif %}
18
19 {% if available_tools %}
20 <available_tools>
21 ## Available Tools
22 {{ available_tools }}
23 </available_tools>
24 {% endif %}
25
26 ## Execution History
27 <history>
28 {{ history }}
29 </history>
30
31 ## Response
32 - if you have completed the task, please output the final report or answer to the user
  question.
33 return in the following format:
34 {
35   "answer": <final answer to the user question, string>,
36   "reason": <reason of the answer>
37 }
38
39 {% if available_tools %}
40 - if you need to extra information, please call a tool, and output the tool name and
  arguments.
41 return in the following format:
42 {
43   "tool_name": <tool name>,
44   "tool_args": <tool arguments>,
45   "thought": <thoughts of the tool call, string>
46 }
47 {% endif %}
48
49 Your answer:

```

FIGURE S14: ReAct Prompt utilized in T-Lens

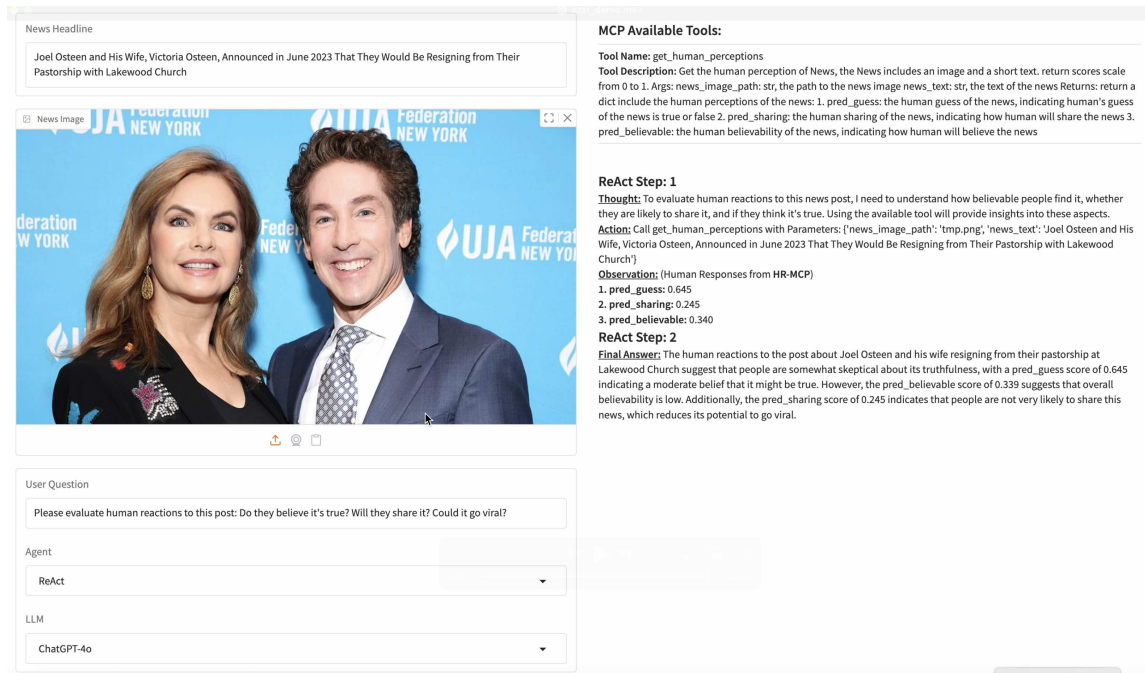


FIGURE S15: A screenshot of T-Lens in predicting human receptivity to a multi-modal post with explanation.

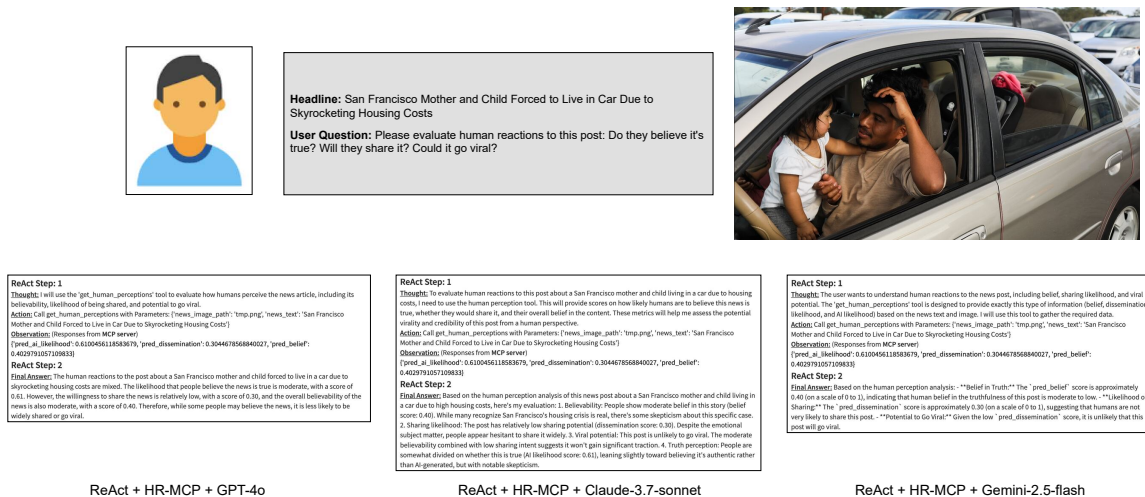


FIGURE S16: Comparing T-Lens' explanations with the responses generated by GPT-4o, Claude-3.7-sonnet and Gemini-2.5-flash in predicting human responses to an online post. HR-MCP can be seamlessly integrated with any LLM.

4 Questionnaire for Our Human Study

Q1. Please let us know how you feel/perceive about the image/video/news/post on the left:

Happiness:

- Not at all
- A little
- A moderate amount
- A lot
- A great deal

Sadness:

- Not at all
- A little
- A moderate amount
- A lot
- A great deal

Anger:

- Not at all
- Extremely slight
- Slight
- Moderate
- Somewhat strong
- Strong
- Extremely strong

Fear:

- Not at all
- Extremely slight
- Slight
- Moderate
- Somewhat strong
- Strong
- Extremely strong

Impressiveness:

- Not at all
- Extremely slight
- Slight
- Moderate
- Somewhat strong
- Strong
- Extremely strong

Q2: How likely will you share this news/post with others, either through verbal communication, re-posting on your social media, or any other means?

- Extremely unlikely
- Very unlikely
- Unlikely
- Neutral
- Likely
- Very likely
- Extremely likely

Q3: How trustworthy is this post? (For social media post)

- Not trustworthy at all
- Highly untrustworthy
- Moderately untrustworthy
- Neutral
- Moderately trustworthy
- Highly trustworthy
- Very trustworthy

Q3': Do you believe this news? (For news)

- Do not believe at all
- Mostly do not believe
- Slightly do not believe
- Neutral
- Slightly believe
- Mostly believe
- Strongly believe

Q4: How likely do you think this news/post is computer generated (i.e., created using computer software or AI algorithms like ChatGPT or GPT-4)?

(For text)

- Extremely unlikely
- Very unlikely
- Unlikely
- Neutral
- Likely
- Very likely
- Extremely likely

Q4': Is this image/video likely computer-generated (e.g., created by AI algorithms like DALL-E, Mid-journey, StableDiffusion) or taken by a camera?

(For visuals)

- Extremely unlikely
- Very unlikely
- Unlikely
- Neutral
- Likely
- Very likely
- Extremely likely

Q5: Please tell us the reasons for your answer in the previous question (select ALL that apply): (For social media text)

- The text is well-written and easy to understand
- The text is poorly-written or hard to understand
- Text style akin to common social media platforms
- Uncommon text style for social media platforms
- Formal style, or usage of uncommon words or specialized terminology
- The text contains much detail

- The text is lack of detail
- The content is inconsistent with known facts
- I have seen similar content before
- Others, please specify: _____

Q5': Please tell us the reasons for your answer in the previous question (select ALL that apply): (For news headline)

- The text is well-written and easy to understand
- The text is poorly-written or hard to understand
- The style is formal or typical of news outlets
- The style is informal or uncommon in news outlets
- Usage of uncommon words or specialized terminology
- The text contains much detail
- The text is lack of detail
- The content seems true to me
- The content seems not true
- I have seen similar news before
- Others, _____ please specify: _____

Q5'': Please tell us the reasons for your answer in the previous question (select ALL that apply): (For visuals)

- The color/lighting looks natural
- The color/lighting looks unnatural
- The objects appear natural
- The objects appear unnatural
- The scene/event depicted in the image/video appears to be unreasonable or fake
- The image/video quality is high
- The image/video quality is low
- I have seen similar image/video before
- The video motion is inconsistent or not smooth
- Others, _____ please specify: _____

Q5''': Please tell us the reasons for your answer in the previous question (select ALL that apply): (For multi-modal news)

- The text is well-written and easy to understand
- The text is poorly-written or hard to understand
- The style is formal or typical of news outlets
- The style is informal or uncommon in news outlets
- Usage of uncommon words or specialized terminology
- The text contains much detail
- The text is lack of detail
- The content seems true to me
- The content seems not true
- I have seen similar news before
- Others, _____ please specify: _____

Q5''': Please tell us the reasons for your answer in the previous question (select ALL that apply):

(For multi-modal social media post)

- The text is well-written and easy to understand
- The text is poorly-written or hard to understand
- The style is formal or typical of news outlets
- The style is informal or uncommon in news outlets
- Usage of uncommon words or specialized terminology
- The text contains much detail
- The text is lack of detail
- The content seems true to me
- The content seems not true
- I have seen similar news before
- Others, _____ please specify: _____

Q6: Rate your agreement with each of the following statements using the scale from 1 to 7, which 1 stands for "Strongly disagree" and 7 means "Strongly agree".

- The content in the image/video matches the content in the text.
- The sentiment of the image/video matches the sentiment of the text.
- The people, objects, or places mentioned in the text are present in the image/video.

Q7: Which genre does this news mainly belong to?

(For multi-modal news)

- Political news
- Health & life science news
- News related to food and food industry
- Entertainment news
- Sports news
- Financial/economic news
- News about disasters and accidents
- News about science (except life science) and technology
- Others

Q7': Which genre does this post belong to? (For multi-modal social media posts)

- User-generated content about daily life — posts involve sharing content created by users themselves
- Memes and humor — content featuring funny images, videos, or captions that relate to popular culture, current events, or trending topics.
- Ads
- Political news
- Health and science news
- News related to food and food industry

- News about education, finance, or sports
- News about disasters and accidents
- Others

5 Limitations

Our human study recruited participants from three countries (the US, India, and Singapore). While these countries represent diverse media environments and cultural backgrounds, findings may not generalize to other regions. Future work should expand to broader geographies.

The Multimodal Sentiment Consistency module in HR-MCP uses simple feature concatenation followed by an MLP, rather than more expressive cross-attention mechanisms (e.g., LXMERT, ALBEF). While MLP fusion is sufficient for our current setting and yields satisfactory performance, cross-attention-based fusion could better capture fine-grained sentiment mismatches between modalities. We leave this as a direction for future work.

6 Additional Related Work

Most prior work on AIGC and misinformation emphasizes automatic detection, but recent studies suggest a broader context. AI tools can improve efficiency, yet they also introduce ethical risks and may reshape scientific review and verification processes, creating an assisted-technology dilemma [22]. Beyond misinformation detection, ML and DL are also increasingly used for human-centered analysis in other domains. A recent scoping review in sports biomechanics shows that these methods are being adopted to model human performance and injury prevention [23]. Prior work reports that Stable Diffusion generated images can be highly realistic and challenging for human observers to distinguish from real ones [?]. This cross-disciplinary trend supports our multimodal design for modeling human responses. It also aligns with recent calls to standardize the assessment of AI and sensor technologies, so systems used in human interaction can be evaluated with rigorous and consistent protocols [24].

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